

Introduction

Housing prices vary significantly across U.S. states due to differences in economic conditions. Understanding these drivers is important for explaining affordability and regional disparities in the housing market.

Literature review

- Prior studies show that housing prices are shaped by income, labor conditions, migration, and cost-of-living differences.
- Regional economic conditions influence affordability and housing demand across states.
- COVID-19 and remote work trends accelerated housing-price growth in many regions.

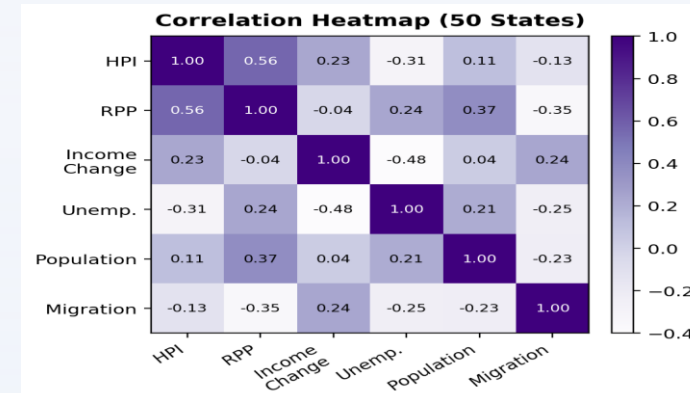
Data collection

- Sources: FHFA, BEA, BLS, Census, IRS
- Time Period: 2020–2024
- Variables:
 - Housing Price Index (HPI)
 - Regional Price Parity (RPP)
 - Income Change
 - Unemployment Rate
 - Net Migration

Data & Method

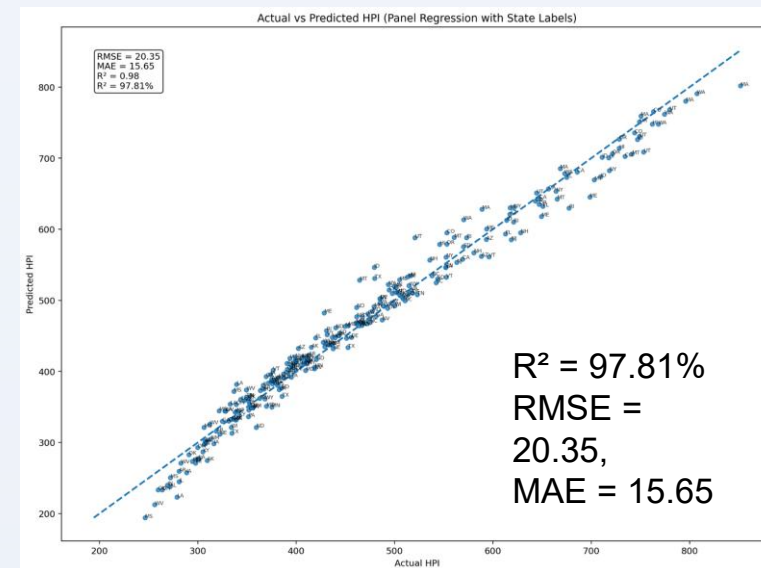
- 2020–2024 50-state panel dataset
- Sources: FHFA, BEA, BLS, Census, IRS
- Main model: two-way fixed effects panel regression
- Comparison model: Random Forest
- State fixed effects control for stable differences across states
- Year fixed effects control for common shocks across years, including COVID-era conditions

Correlation Heatmap

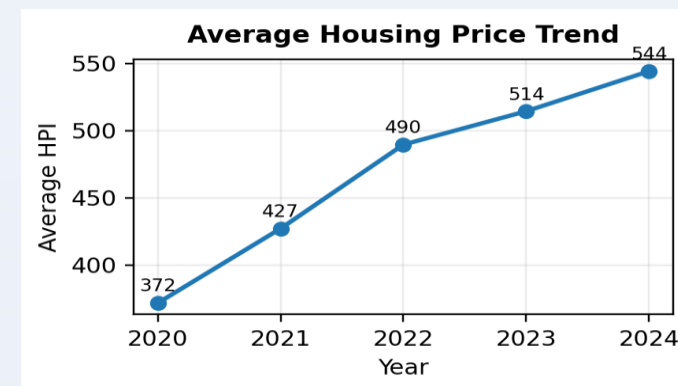


RPP shows the strongest relationship with HPI; migration is the weakest.

Actual vs Predicted HPI

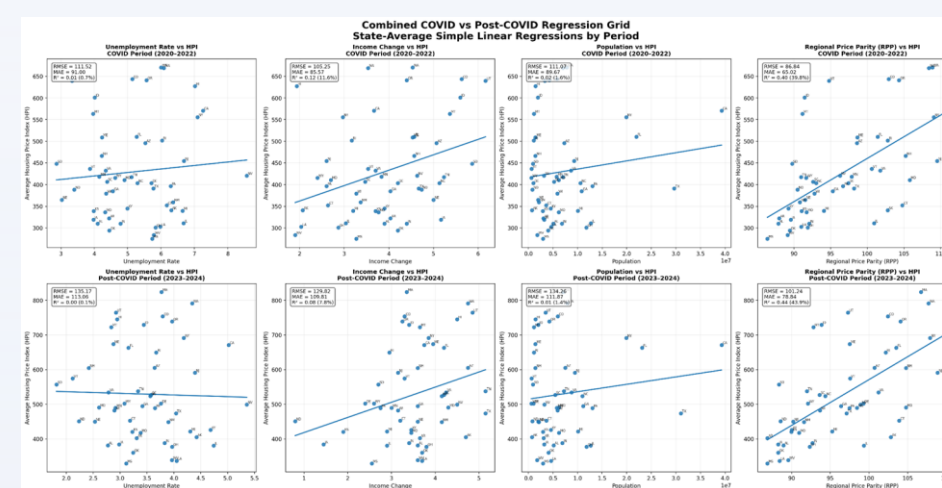


Housing Price Trend

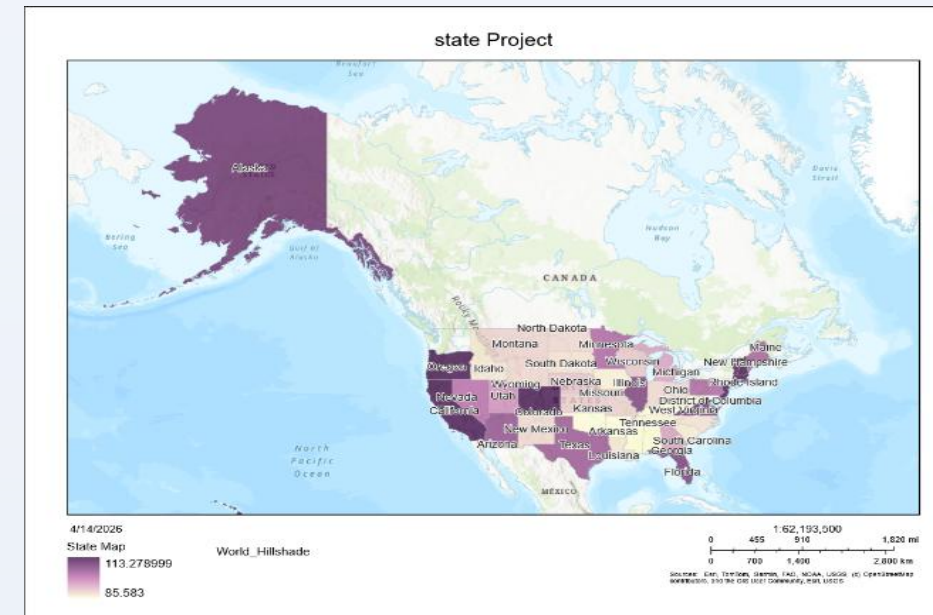


The average HPI rose from 372 in 2020 to 544 in 2024.

COVID vs Post-COVID Panel Regression Comparison



Spatial Analysis



Choropleth map of housing price index across U.S. states. Darker shades indicate higher housing price levels, highlighting regional disparities with higher concentrations in coastal and economically strong states.

Model Concept:

Panel Regression Model:

$$\text{HPI} = \beta_0 + \beta_1(\text{RPP}) + \beta_2(\text{Income Change}) + \beta_3(\text{Unemployment}) + \beta_4(\text{Population}) + \text{State FE} + \text{Year FE} + \epsilon$$

Migration was analyzed as a supporting variable for the reduced 2021–2023 subset because IRS migration data do not cover the full panel.

Results (Main Findings)

- RPP was one of the strongest and most consistent factors associated with housing prices
- Unemployment remained negatively associated with housing prices
- Income contributed to the broader housing-price story
- Population was weaker than the main labor and cost-of-living variables
- Migration was more limited because of shorter data coverage
- The panel regression fit the data strongly: R² = 97.81%, RMSE = 20.35, MAE = 15.65

Interpretation

- Housing prices are best explained through a multivariable framework rather than a single economic factor. In the 50-state panel, cost-of-living differences (RPP) emerged as one of the strongest and most consistent factors, while unemployment and income provided additional explanatory power over time.

Conclusion

- Housing prices are best explained by multiple interacting economic factors
- RPP / cost of living remained one of the clearest signals across the 50-state panel
- The two-way fixed effects panel regression better matched the structure of the data
- Next steps: add housing supply, interest rates, mortgage availability, and fuller migration coverage

References

- [1] U.S. Census Bureau, “State Population Totals and Components of Change: 2020–2025,” 2026.
- [2] U.S. Bureau of Labor Statistics, “Local Area Unemployment Statistics (LAUS),” 2025.
- [3] Federal Housing Finance Agency, “FHFA House Price Index Datasets,” 2025.
- [4] U.S. Bureau of Economic Analysis, “Regional Price Parities by State and Metro Area,” 2024.
- [5] E. L. Glaeser, J. Gyourko, and R. E. Saks, “Why Have Housing Prices Gone Up?” American Economic Review, vol. 95, no. 2, pp. 329–333, May 2005..